

Predicting Depression Among University Students Using Machine Learning

1. Introduction

Mental health issues, especially students' mental health issues, are always a highly concerning problem among all universities. In many campuses—UCLA, for example—students frequently experience overwhelming stress, loneliness, or academic difficulties that can escalate into serious conditions such as depression. Depression can negatively impact students' academic performance, mental and physical health, and long-term development, including but not limited to the inability to complete coursework, self-destructive behavior, and even suicide. Therefore, early identification of high-risk students is an extremely important public health objective.

Within this context, this project aims to statistically predict whether students will report depressive symptoms using demographic, academic, and psychological variables collected through questionnaires. By constructing and comparing multiple machine learning models, including logistic regression, random forest, and MLP, this study aims to explore which factors are most strongly associated with depression and to evaluate the predictive value of various indicators, especially psychological indicators. In the practical application section, our goal is for this machine learning project to help universities identify students who may require early mental health intervention.

2. Data

This project uses a dataset from a survey on university student mental health on Kaggle. The dataset contains approximately one hundred observations, covering variables such as demographic information, academic background, and mental health status.

The predictors span three conceptual groups:

1. Demographic variables: Gender, Age, Marital status
2. Academic variables: Course of study, Year of study, CGPA
3. Psychological indicators: Anxiety, Panic attacks, Treatment-seeking behavior

The dataset is clean, well-organized, and clearly structured, with no difficult-to-understand parts. However, there are some significant limitations:

- (1) The sample size is small.
 - (2) The responses are self-reported, which may introduce bias.
 - (3) The data comes from a single university (International Islamic University Malaysia), limiting its generalizability.
 - (4) Mental health variables may be sensitive and prone to misreporting or underreporting.
- The target variable is Depression, which indicates whether a student self-reported experiencing symptoms of depression (“Yes” or “No”).

3. Preprocessing

Although the dataset is complete and clear, several preprocessing steps are still necessary to prepare it for modeling. This ensures that all variables are clean and numerical, while also reducing interference caused by issues such as missing data.

3.1. General data cleaning, missing values, and categorical variable conversion

Column names were simplified, and all string variables were standardized for spacing and capitalization to avoid issues such as 'year' and 'Year' not being recognized as the same. Most categorical variables were converted to binary numerical form:

Gender $\rightarrow \{0, 1\}$, 0 for male, 1 for female

Marital status $\rightarrow \{0, 1\}$, 0 for no, 1 for yes

Anxiety, panic disorder, seeking treatment $\rightarrow \{0, 1\}$, same as above

Depression $\rightarrow \{0, 1\}$, same as above, used as the target variable

This transformation enabled the use of machine learning models, which require numeric input.

Additionally, one row contained a missing value in the "Age" column. Due to the small size of the dataset, interpolation might introduce unnecessary bias, so this row was removed to prevent missing value errors during cross-validation.

```
# preprocessing

# adjust column names
df.rename(columns={
    'What is your course?': 'Course',
    'Your current year of Study': 'Year',
    'What is your CGPA?': 'CGPA',
    'Marital status': 'Marital',
    'Do you have Anxiety?': 'Anxiety',
    'Do you have Panic attack?': 'Panic',
    'Did you seek any specialist for a treatment?': 'Treatment'
}, inplace=True)

# Year
df['Year'] = (
    df['Year']
    .str.lower()          # 'Year 1' -> 'year 1', etc
    .str.extract(r'(\d+)')
    .astype(float)
)

# numericalize all yes/no answers
yes_no_cols = ['Depression', 'Anxiety', 'Panic', 'Marital', 'Treatment']

for col in yes_no_cols:
    df[col] = df[col].str.strip().str.title()    # 'yes ' -> 'Yes'
    df[col] = df[col].map({'No': 0, 'Yes': 1})

# numericalize gender
df['Gender'] = df['Gender'].str.strip().str.title()
df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})
```

Figure 1: The code used for data cleaning

```
df[df['Age'].isna()]
```

	Timestamp	Gender	Age	Course	Year	CGPA	Marital	Depression	Anxiety	Panic	Treatment	Course_clean	Course_STEM	CGPA_num
43	8/7/2020 15:07	0	NaN	BIT	1.0	0 - 1.99	0	0	0	0	0	Bit	1	0.995

Figure 2: The line dropped

3.2. Course simplification: STEM vs non-STEM

The "Course" variable initially contained many unique and unordered categories (Engineering, Islamic Education, BIT, Law, etc.). These categories are easy for humans to understand but meaningless to machines. Further, these subject names have no inherent order, and label encoding them would implicitly assign a meaning to them that would be captured by models like random forests, but which doesn't actually represent meaningful numerical distances. To prevent this implicit error, all courses were manually grouped into STEM and non-STEM based on their subject area. A new binary feature, "Course_STEM," was created. By abstracting the specific subject names into STEM and non-STEM, the new Course_STEM factor retains interpretability for machines and prevents misleading ordinal relationships. (Thanks to Dr. Alexander Lozinski for suggesting this approach!)

```
df['Course_clean'] = df['Course'].str.strip().str.title()
df['Course_clean'].unique()

array(['Engineering', 'Islamic Education', 'Bit', 'Laws', 'Mathemathics',
       'Pendidikan Islam', 'Bcs', 'Human Resources', 'Irkhs',
       'Psychology', 'Kenms', 'Accounting', 'Enm', 'Marine Science',
       'Koe', 'Banking Studies', 'Business Administration', 'Law',
       'Kirkhs', 'Usuluddin', 'Taasl', 'Engine', 'Ala',
       'Biomedical Science', 'Benl', 'It', 'Cts', 'Engin', 'Econs',
       'Mhsc', 'Malcom', 'Kop', 'Human Sciences', 'Biotechnology',
       'Communication', 'Diploma Nursing', 'Radiography', 'Fiqh Fatwa',
       'Diploma Tesl', 'Fiqh', 'Nursing'], dtype=object)
```

Figure 3: all existing majors in the dataset

```
# STEM and non-STEM sets based on manual classification
stem_set = {
    'Engineering', 'Bit', 'It', 'Engine', 'Engin',
    'Marine Science', 'Biomedical Science', 'Biotechnology',
    'Radiography', 'Nursing', 'Diploma Nursing',
    'Mathemathics', # typo for Mathematics
    'Mhsc', # Faculty of Health Sciences (IIUM)
    'Koe' # Faculty of Engineering (IIUM)
}

non_stem_set = {
    'Islamic Education', 'Pendidikan Islam', 'Usuluddin', 'Fiqh', 'Fiqh Fatwa',
    'Taasl', 'Benl', 'Laws', 'Law',
    'Business Administration', 'Accounting', 'Banking Studies',
    'Econs', 'Human Resources', 'Human Sciences',
    'Irkhs', 'Kirkhs', 'Kenms', 'Enm', 'Cts',
    'Malcom', 'Kop', 'Communication',
    'Ala', 'Bcs' # conservative decision for ambiguity
}

# Function to classify
def classify_course(course):
    if course in stem_set:
        return 1
    elif course in non_stem_set:
        return 0
    else:
        # safety fallback (never happens if classification list is complete)
        return 0

df['Course_STEM'] = df['Course_clean'].apply(classify_course)
```

Figure 4: Implementation of course simplification. 1 for STEM majors, 0 for non-STEM majors

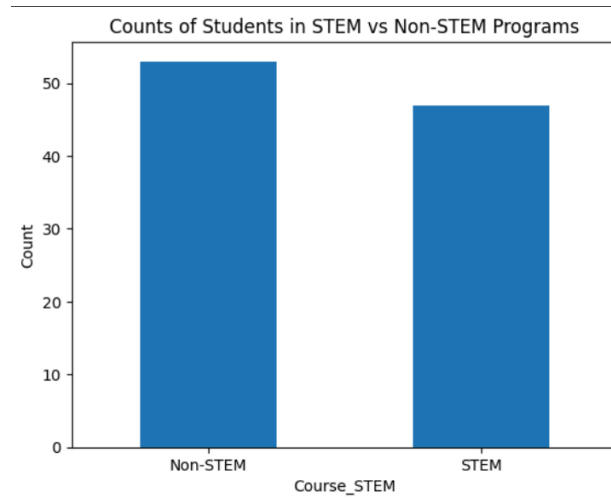


Figure 5: Visualizing the simplified Course_STEM

3.3. CGPA interval numeric transformation

The CGPA values in the original dataset were provided in the form of ranges, such as "3.00–3.49," which led to similar problems as with the course data: it was difficult to input this type of data into a machine learning model, and the model would struggle to identify the insights within it. The initial attempt was to categorize each range, such as labeling GPA 0-1.99 as 0, 2-2.49 as 1, etc. The advantage of this approach was that it preserved the classification ability of the original format and its implied numerical distance. However, this introduced a problem: the scale of the categorical values 0, 1, 2 is different from the scale of the ranges 0-1.99, 2-2.49, 2.5-2.99, which have specific meanings. Even if these two were mapped one-to-one, the impact on the model's recognition ability would still exist. Therefore, an improved method was applied: converting each range to its numerical midpoint. This method avoids arbitrary assumptions about the distribution within the range, while preserving the order and not changing the data scale, making CGPA suitable for quantitative modeling.

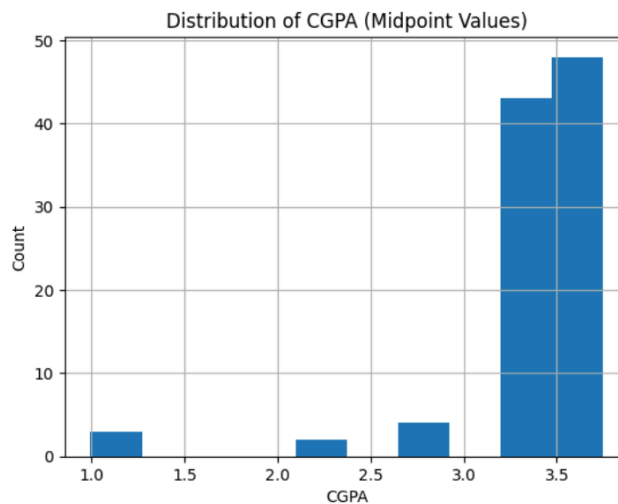


Figure 6: A briefing of the CGPA factor after transformation

3.4. Feature sets: Basic vs Full

The features were categorized into two sets, basic and full, for training. The basic feature set will be used to explore the accuracy of predicting depression status without relying on other mental health indicators, while the full feature set will include three additional psychological indicators: Anxiety, Panic, and Treatment. Anxiety and panic sometimes lead to depression, and sometimes are comorbid symptoms of depression; whether or not a person receives treatment is an important indicator of overall mental health. The training results of the two feature sets will be compared to determine the predictive value of the psychological indicators:

Basic feature set: Gender, Age, Year, Course_STEM, CGPA_num, Marital

Full feature set: Basic features + Anxiety, Panic, Treatment

This comparison can isolate the additional predictive power contributed by the mental health variables and help to gain a more detailed understanding of the factors associated with depression.

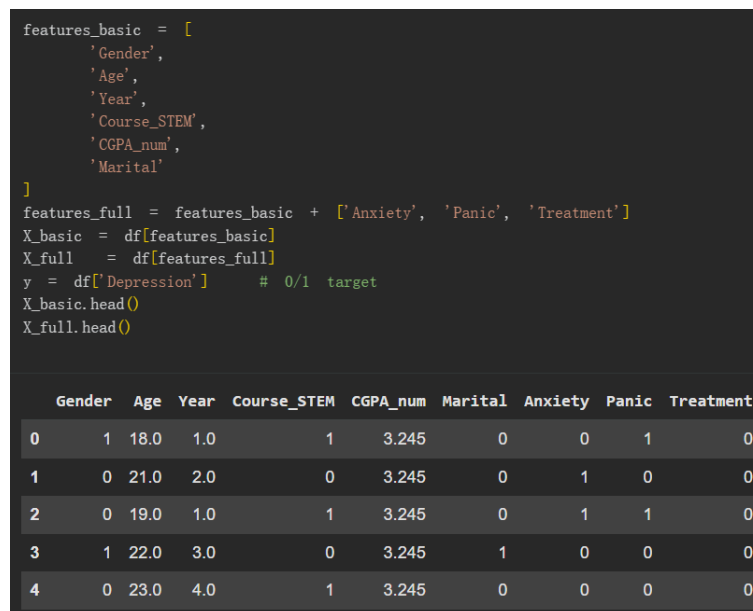


Figure 7: A briefing of the feature sets

4. Modeling

To ensure the reliability of the conclusions, this project trained and compared three different machine learning models to predict whether students would report experiencing depression. The three models—logistic regression, random forest, and MLP—represent different families of classifiers: linear models, tree models, and neural network models. This ensures the generalizability of the project, allowing for a comprehensive comparison and selection of the optimal model while covering as wide a range of possibilities as possible. Also, a 5-fold cross-validation was applied to each model to ensure a similar proportion of depressed and non-depressed students in each fold. The importance of each factor in each model is also listed.

4.1. Logistic Regression (LR)

Logistic regression was chosen as the base model for this project due to its simplicity, interpretability, and suitability for small datasets. The advantage of linear models in quickly adapting to relatively small amounts of data is particularly evident and thus makes the model perform well.

Logistic regression assumes a linear relationship between the predictor variables and the log-odds of depression, which allows the influence of each feature to be directly understood through its coefficient.

With a random state of 27, the logistic regression model using the basic feature set showed an accuracy of 81%, while the model using the complete feature set showed a slight improvement, achieving an accuracy of 82%. For logistic regression, the most contributing and important factors were Marital status, Anxiety, and Treatment.

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold, cross_val_score
import numpy as np

# Logistic Regression model
log_reg = LogisticRegression(max_iter=2000)

# 5-fold stratified cross validation
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=27)

# Evaluate basic model
scores_basic = cross_val_score(log_reg, X_basic, y, cv=cv, scoring='accuracy')
scores_full = cross_val_score(log_reg, X_full, y, cv=cv, scoring='accuracy')

print("Basic Model Accuracy: ", scores_basic, "Mean:", scores_basic.mean())
print("Full Model Accuracy: ", scores_full, "Mean:", scores_full.mean())

Basic Model Accuracy: [0.8 0.8 0.75 0.9 0.8 ] Mean: 0.8099999999999999
Full Model Accuracy: [0.85 0.65 0.85 0.9 0.85] Mean: 0.82
```

Figure 8: Logistic Regression

```
# check each factor's importance
log_reg.fit(X_full, y)

coef = log_reg.coef_[0] # coefficient vector

for name, c in zip(X_full.columns, coef):
    print(f"{name:15s} {c:.4f}")
```

Gender	0.6414
Age	-0.0932
Year	-0.2645
Course_STEM	-0.1371
CGPA_num	-0.2236
Marital	2.6910
Anxiety	1.0099
Panic	0.4902
Treatment	0.9700

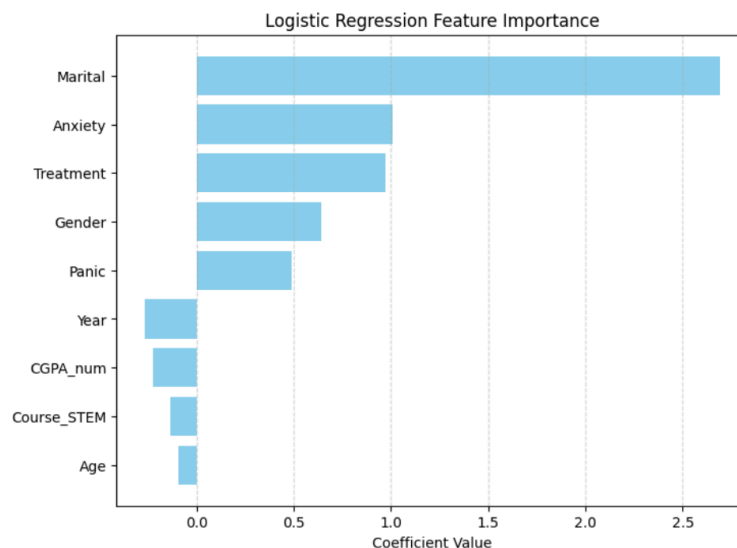


Figure 9&10: Feature importance in LR model

4.2. Random Forest (RF)

The random forest was chosen as a typical non-linear model to compare against linear models. A random forest is a large ensemble model composed of multiple (potentially "very many," depending on the requirements) decision trees, capturing non-linear relationships and feature interactions through recursive binary splitting.

The biggest difference from logistic regression is that random forests do not assume a linear relationship and can discover threshold effects (e.g., specific GPA or age thresholds). This is especially important in situations where psychological factors, which cannot be easily quantified and may interact with other elements in complex ways, are present, as such relationships are extremely difficult to capture with linear models. What's more, a characteristic of "black box" models such as deep neural networks and random forests is that, with the correct parameters, the model quality improves at an astonishing rate.

With a `random_state = 27`, the random forest benefited significantly from the complete feature set, achieving the best performance among all models with an accuracy of 86%. However, this was not achievable with the industry standard state, i.e., `random_state = 42`. The random forest model with `random_state = 42` did not perform better than LR. This fully demonstrates the value of selecting RF as a comparison model and the potential of this model.

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score

rf = RandomForestClassifier(
    n_estimators=300,
    max_depth=None,
    min_samples_split=2,
    random_state=27
)

# 5-fold CV
scores_rf_basic = cross_val_score(rf, X_basic, y, cv=cv, scoring='accuracy')
scores_rf_full = cross_val_score(rf, X_full, y, cv=cv, scoring='accuracy')

print("RF Basic acc:", scores_rf_basic, "mean =", scores_rf_basic.mean())
print("RF Full acc:", scores_rf_full, "mean =", scores_rf_full.mean())

RF Basic acc: [0.75 0.65 0.7 0.95 0.8 ] mean = 0.7699999999999999
RF Full acc: [0.85 0.8 0.85 0.95 0.85] mean = 0.86
```

Figure 11: Random Forest

```
# check each factor's importance
rf.fit(X_full, y)

importances = rf.feature_importances_

for name, score in zip(X_full.columns, importances):
    print(f"{name:15s} {score:.4f}")

Gender          0.0572
Age             0.1757
Year            0.0940
Course_STEM     0.0546
CGPA_num        0.0957
Marital         0.2953
Anxiety         0.0910
Panic           0.0611
Treatment       0.0754
```

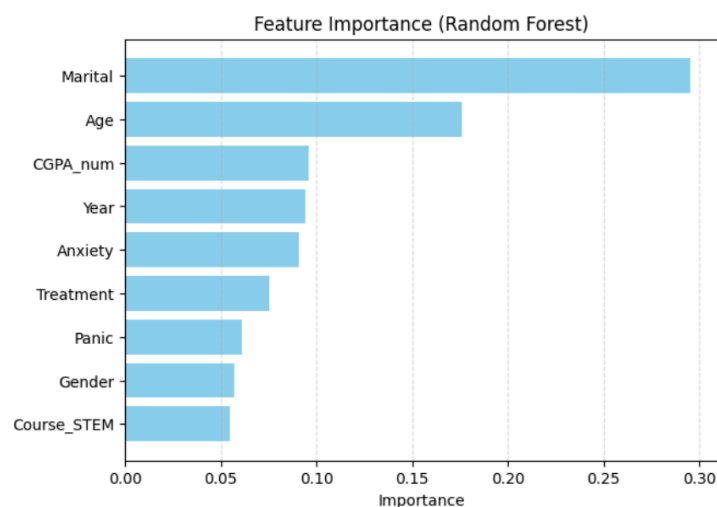


Figure 12&13: Feature importance in RF model

4.3. Random Forest (RF)

The Multi-Layer Perceptron (MLP) was included as a representative non-linear model. In the long run, neural network models tend to learn deep and complex interactions. A major disadvantage of neural networks is that they typically require larger datasets to generalize well (in fact, very small datasets can cause neural networks to perform extremely poorly).

However, our dataset was not so small that the neural network couldn't learn anything, and including a neural network model allowed us to test whether a deeper model could surpass the predictive performance of logistic regression and random forests.

However, no miracle occurred. As expected, the small size of the dataset (approximately 100 samples) resulted in stable, moderate, but not outstanding performance for the MLP, and it did not benefit significantly from the additional psychological features like the other models.

```
from sklearn.neural_network import MLPClassifier

mlp = MLPClassifier(
    hidden_layer_sizes=(16, 8),
    activation='relu',
    solver='adam',
    max_iter=3000,
    random_state=27
)

# 5-fold CV
scores_mlp_basic = cross_val_score(mlp, X_basic, y, cv=cv, scoring='accuracy')
scores_mlp_full = cross_val_score(mlp, X_full, y, cv=cv, scoring='accuracy')

print("MLP Basic acc:", scores_mlp_basic, "mean =", scores_mlp_basic.mean())
print("MLP Full acc:", scores_mlp_full, "mean =", scores_mlp_full.mean())

MLP Basic acc: [0.8 0.65 0.75 0.9 0.8 ] mean = 0.78
MLP Full acc: [0.85 0.6 0.75 0.85 0.85] mean = 0.78
```

Figure 14: MLP Classifier

```
# check each factor's importance
from sklearn.inspection import permutation_importance

mlp.fit(X_full, y)

# permutation importance
result = permutation_importance(
    mlp, X_full, y,
    n_repeats=30,
    random_state=27,
    scoring='accuracy'
)

importances = result.importances_mean

for name, score in zip(X_full.columns, importances):
    print(f"{name:15s} {score:.4f}")

Gender          0.0077
Age             -0.0123
Year            -0.0033
Course_STEM     -0.0057
CGPA_num        0.0077
Marital         0.1610
Anxiety         0.0397
Panic           -0.0003
Treatment       0.0297
```

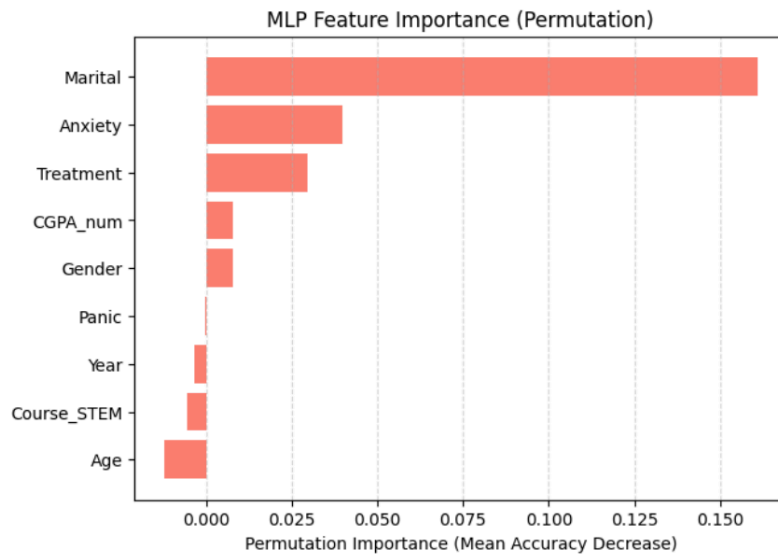


Figure 15&16: Feature importance in MLP model

5. Results

The specific implementation methods of three machine learning models—logistic regression (LR), random forest (RF), and multilayer perceptron (MLP)—have been demonstrated. In all experiments, performance was evaluated using 5-fold stratified cross-validation to ensure stable estimation results even with a limited sample size.

5.1. Model Performance Comparison

Model	Basic	Full
Logistic Regression	0.81	0.82
Random Forest	0.77	0.86
MLP Neural Network	0.78	0.78

Table 1: Model Comparison

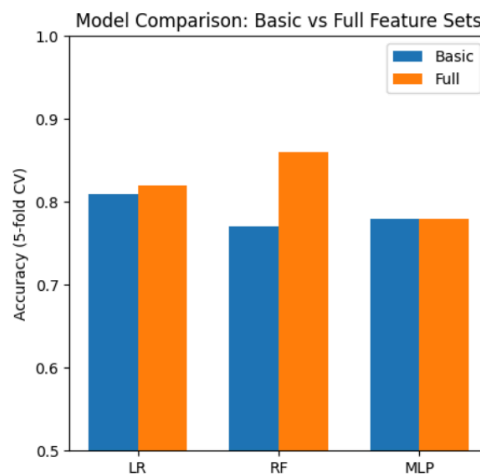


Figure 17: Model Comparison Visualization

It can be observed that:

The performance of the random forest model improved most significantly after adding psychological variables (anxiety, panic, treatment), increasing from 0.77 to 0.86. This propelled the random forest model from being the worst-performing model to the strongest model with a dramatic improvement in performance.

The performance of the logistic regression model improved slightly (+0.01), indicating that these variables still have some predictive value from a linear perspective.

The performance of the MLP model did not improve, which is most likely due to the small size of the dataset—neural networks typically require larger training samples.

From this, we can draw a preliminary conclusion:

Do academic, demographic, and psychological factors jointly improve the ability to predict depression?

The answer is yes, especially for models that can learn non-linear interactions (such as random forests). However, for models that rely on depth and data volume, such as neural networks, we need more data for training to obtain further conclusions.

5.2. Feature Importance Across Models

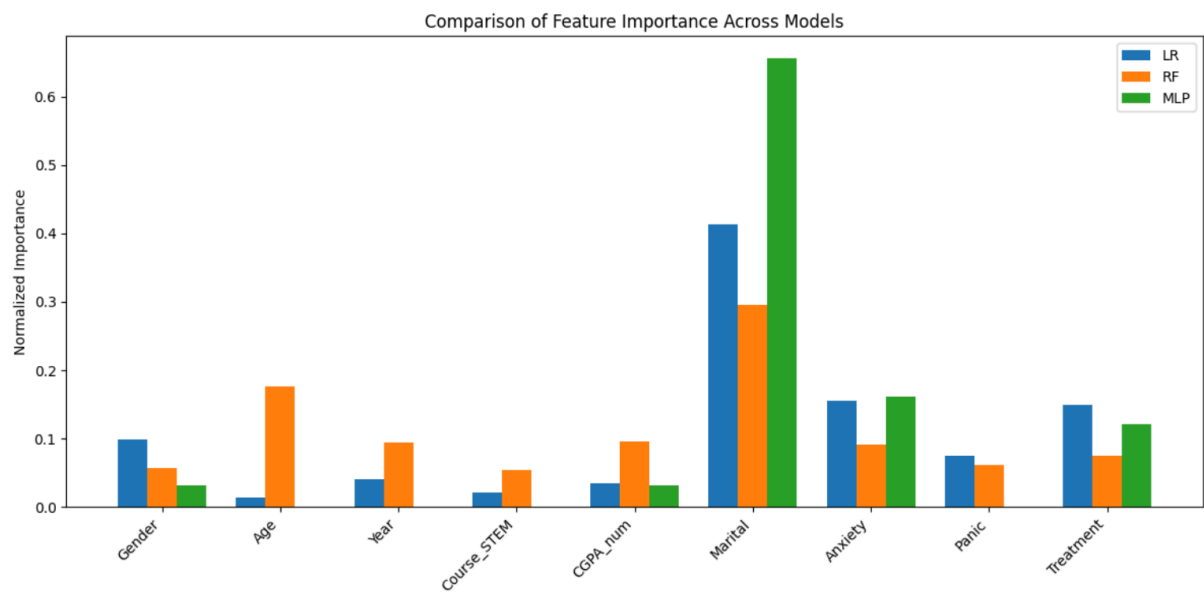


Figure 18: Feature Importance Across Models

As is evident from Figure 18, the most important predictors of self-reported depression across all three models are:

1. Marital status. This is the most important coefficient by a significant margin. LR assigned the largest positive coefficient, RF gave the highest importance score, and MLP showed the

largest permutation importance drop. This indicates extreme robustness: marital status is the most consistent indicator of depression risk in this dataset. From the LR perspective, being married is highly likely to lead to depression.

2. Psychological variables (anxiety, panic, treatment), as the main focus of this project, also showed some importance. LR highly valued anxiety and treatment, RF considered anxiety moderately important, while panic/treatment had lower importance, and MLP assigned positive importance to both anxiety and treatment.

Psychological self-assessments significantly contribute to prediction, but their relative weights differ across different model types. This is due to the different ways in which the models learn.

3. Academic and demographic variables varied in importance. RF valued age, year, and cumulative grade point average (CGPA) more than LR or MLP, while LR found a small negative linear association with these variables (older students and students with higher CGPA were slightly less likely to report depression). MLP generally considered these variables to be of lower importance. Overall, the importance of these variables is not as high as the first two categories.

From this, we can draw a less technical conclusion:

Most important features:

- Marital status
- Anxiety
- Treatment

Moderate importance:

- Age, Year, CGPA (RF > LR/MLP)

Low or inconsistent importance:

- Gender, Panic, Course_STEM

6. Discussion

6.1 The dominant role of marital/relationship status

In all three models, "marriage" was the most important single predictor of depression.

Logistic regression assigned it the largest positive coefficient, while both random forest and multilayer perceptron models ranked it as the most important feature. It's worth noting that in this dataset, "yes" indicates having a partner. Therefore, a positive logistic regression coefficient means that:

Students in relationships are more likely to report depression than single students.

This is, in fact, a completely unexpected conclusion. However, while it may seem counterintuitive at first glance, it can be reasonably explained. For example, conflict or instability in a romantic relationship can cause emotional stress. Students with pre-existing mental health issues may be more inclined to seek emotional support through interpersonal

relationships. Also, due to the small sample size and the self-reported nature of the data, this may not be a general trend. Yet, the fact that three distinctly different models consistently identified this factor as the most important suggests that this conclusion is meaningful and should not be ignored.

6.2. Psychological Variables

A core objective of this study was to examine whether psychological variables could improve predictive power. The results clearly support this objective: logistic regression showed a slight improvement (+0.01) when using the complete feature set, the random forest model showed a significant improvement (+0.09), and while the MLP's performance did not improve, its feature dependence shifted significantly towards psychological variables. This suggests that psychological indicators—especially anxiety and treatment status—provide meaningful predictive information beyond demographics and academic performance.

It is worth noting that the degree of improvement varied across different models, as different model types capture different relationships. Logistic regression detects linear associations (e.g., higher anxiety levels lead to a higher probability of depression); random forests (RF) detect non-linear thresholds or interactions (e.g., anxiety interacts with age or cumulative grade point average (CGPA)); and multilayer perceptrons (MLPs) can learn complex non-linear combinations, although, as emphasized above, they are limited by the small sample size. Notably, complex models often perform worse than simpler models when the dataset is small.

Anxiety and treatment showed significant and consistent importance across all models, aligning with common sense and clinically relevant academic understanding: students with anxiety symptoms or who have sought treatment are more likely to also suffer from depression, as depression and anxiety can mutually reinforce each other, and treatment is an indicator of overall mental health status.

6.3. Limitations

The following limitations should be noted:

1. Small Sample Size

This limitation is repeatedly mentioned throughout the report. This dataset only contains approximately 100 observations, which somewhat limits the statistical power and generalizability of the models. The surprisingly high importance of marital status may also be due to this. However, this limitation mainly affected the MLP; LR and RF were still able to learn a decent amount of information from this dataset.

2. Self-Reported Data

This dataset is self-reported, and self-reporting data, especially mental health data such as depression, anxiety, and panic attacks, is not always accurate. This may introduce bias and underreporting. For example, there may be cases where individuals are depressed but unaware of it, or unwilling to let others know about their mental state. Introducing clinical assessments would significantly improve the interpretability of this dataset.

3. Lack of Temporal Dynamics

All data comes from the same point in time, so causal conclusions cannot be drawn. The relationships may reflect correlation rather than underlying causes. To draw more precise and specific conclusions, longer-term tracking to obtain longitudinal data would be beneficial.

7. Conclusion

The goal of this project was to identify and predict depression and depressive tendencies in university students, exploring which factors provide the most significant predictive value for identifying depression in university students, with a concentration on psychological factors. The study used three different machine learning models (logistic regression, random forest, and MLP), and the results consistently showed that psychological variables improved predictive accuracy, with the most significant improvement observed in the random forest model.

In all models, marital status was the most important factor in predicting depression. In this dataset, university students in a romantic relationship were more likely to report depression. This was an unexpected and extremely important finding. This pattern reflects the immense pressure that interpersonal stress and emotional burden can place on individuals. Anxiety and treatment history were also important predictors, illustrating the interconnectedness of mental health symptoms.

Comparison of different model types showed that different learning mechanisms—linear associations in logistic regression, non-linear interactions in random forests, and flexible representation learning in MLP—were able to capture different aspects of the data. Despite these differences, all models arrived at a common conclusion: psychological indicators contain meaningful and undeniable information.

The limitations of this study include a small sample size and self-reported data. The findings should be considered exploratory rather than conclusive. However, the direction of this research appears to be correct: the consistency of conclusions across different modeling methods suggests that the patterns discovered warrant further, longitudinal research in larger and more diverse student populations.

Combining psychological variables with demographic and academic information can lead to more accurate and valuable predictions. These findings can inform screening efforts for students at risk of depression. From a preliminary application perspective, universities should prioritize students who are married or in a committed relationship, as well as those who have already experienced some mental health issues, as they have a higher probability of depression.

8. References

“Student Mental Health.” *W*www.kaggle.com,

www.kaggle.com/datasets/shariful07/student-mental-health/data.